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Edwards' syndrome is a condition resulting from an extra chromosome 18. A chromosome mutation in the second meiotic division is the most frequent cause of Edwards' syndrome.

Explain how a chromosome mutation in the second meiotic division could result in an extra chromosome 18.

In your answer, **name** the type of chromosome mutation which would result in the extra chromosome.

[2 marks]

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Complete trisomy 18 is the most common type of Edwards' syndrome. This occurs when all the cells of the body have an extra chromosome 18.

Explain why all the cells of the body have an extra chromosome 18.

[2 marks]



0 7 . 3

Mosaic trisomy 18 is another type of Edwards' syndrome. This occurs due to a chromosome mutation after fertilisation.

In mosaic trisomy, the body has cells with an extra chromosome 18 **and** cells with the correct number of chromosomes.

Explain how cells with different numbers of chromosomes are produced in mosaic trisomy.

[1 mark]

0 7 . 4

The age of the female parent is a factor linked to the risk of a child having Edwards' syndrome.

Which statistical test should be used to test whether this link is statistically significant?

Tick (✓) **one** box.

[1 mark]

Correlation coefficient

☐

Chi-squared

☐

Student's t-test

☐

Question 7 continues on the next page

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A ventricular septal defect (VSD) is a common feature of Edwards' syndrome.

A VSD is a hole in the wall between the two ventricles of the heart. A VSD can cause higher blood pressure in the lungs.

Explain how a VSD can cause higher blood pressure in the lungs.

[2 marks]

8

